

Expectation Decider

Probability Analysis of Student Exam Results

Project	PR-1: Mathematics and Advanced Statistics
Dataset	student_data.csv (200 students, AI-generated)
Tools	Python, pandas, matplotlib, scipy
Companion file	Expectation_Decider.ipynb (code and outputs)

About the data. The 200 student records were generated with an AI tool (Python) so that every task in the project can be solved. Columns: *study_hours*, *attendance*, *group_discussion*, *previous_test_score*, *final_exam_pass*.

Contents

- Task 1: Understanding the basics
- Task 2: Types of events
- Task 3: Random variable and probability distribution
- Task 4: Venn diagram
- Task 5: Contingency table and probability calculations
- Task 6: Understanding relationships
- Task 7: Bayes theorem
- Final summary

Task 1: Understanding the Basics

What is Probability?

Probability tells us how likely it is that something will happen. It is a number from 0 to 1. A value of 0 means the event can never happen, 1 means it will surely happen, and 0.5 means it is equally likely to happen or not.

$$P(E) = (\text{number of favourable outcomes}) / (\text{total number of outcomes})$$

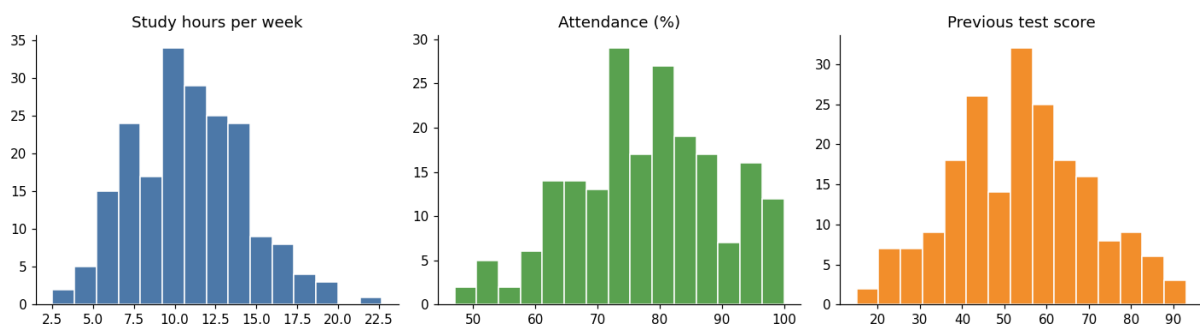
Key Probability Terms

Term	Simple meaning	Example from our data
Experiment	An action with an uncertain result	Picking one student at random
Outcome	One possible result	The chosen student passed
Sample space (S)	All possible outcomes	All 200 students
Event (E)	A group of outcomes we care about	Students who passed
Complement (E')	E does not happen	Students who failed
Mutually exclusive	Cannot happen together	Pass and Fail
Independent events	One event does not change the chance of the other	Tested in Task 6
Joint probability	Both A and B happen: $P(A \text{ and } B)$	Group discussion and Pass
Marginal probability	Chance of one event alone	$P(\text{Pass})$
Conditional probability	Chance of A when B is known: $P(A B)$	$P(\text{Pass} \text{group discussion})$

Three Probability Event Examples from the Dataset

Event	Favourable	Total	Probability
Student passes the exam	91	200	$91/200 = 0.455$
Study hours > 10 per week	115	200	$115/200 = 0.575$
Attendance > 80%	82	200	$82/200 = 0.410$

Distribution of the three numeric columns:



Task 2: Types of Events

Empirical probability comes from real, observed data: $P = (\text{times the event happened}) / (\text{total trials})$.

Theoretical probability comes from logic, when all outcomes are equally likely: $P = (\text{favourable outcomes}) / (\text{all possible outcomes})$.

Empirical scenario: a random student passes the exam

$P(\text{Pass}) = \text{students who passed} / \text{total students}$

$P(\text{Pass}) = 91 / 200 = 0.4550$

Theoretical scenario: a random student takes part in group discussion

There are 2 possible outcomes (Yes or No). If both are equally likely, then:

$P(\text{Yes}) = 1 / 2 = 0.5000$

Comparison: in the real data, $P(\text{Yes}) = 104/200 = 0.5200$. The difference is only +0.02, so the equal-chance idea is a good guess here. For passing, we cannot use logic because the chance of passing is not 50-50, so the empirical value is the one we trust.

Task 3: Random Variable and Probability Distribution

Random variable: a rule that gives a number to the result of a random experiment.

X = number of students who pass the exam out of 3 randomly selected students. X can be 0, 1, 2 or 3.

Each student passes with $p = 0.455$ and fails with $q = 1 - p = 0.545$. Treating the 3 students as independent trials, X follows a Binomial distribution with $n = 3$.

$$P(X = x) = C(3, x) * p^x * q^{(3-x)}$$

Probability Distribution Table

x	Working	P(X = x)	x * P(x)	x ² * P(x)
0	$C(3,0) * 0.455^0 * 0.545^3$	0.1619	0.0000	0.0000
1	$C(3,1) * 0.455^1 * 0.545^2$	0.4054	0.4054	0.4054
2	$C(3,2) * 0.455^2 * 0.545^1$	0.3385	0.6770	1.3539
3	$C(3,3) * 0.455^3 * 0.545^0$	0.0942	0.2826	0.8478
Total		1.0000	1.3650	2.6071

Mean and Variance

Mean: $E(X) = \text{sum of } x * P(x) = 1.3650$

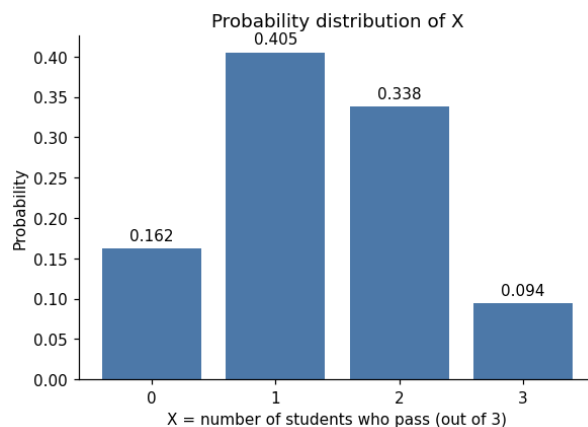
$E(X^2) = \text{sum of } x^2 * P(x) = 2.6071$

Variance: $\text{Var}(X) = E(X^2) - [E(X)]^2 = 2.6071 - 1.8632 = 0.7439$

Std. deviation = $\sqrt{0.7439} = 0.8625$

Check with shortcut formulas: $n * p = 3 * 0.455 = 1.3650$ and $n * p * q = 3 * 0.455 * 0.545 = 0.7439$. Both match.

Meaning: If we pick 3 students at random, we expect about 1.37 students to pass. Exactly 1 pass is the most likely result (0.405).



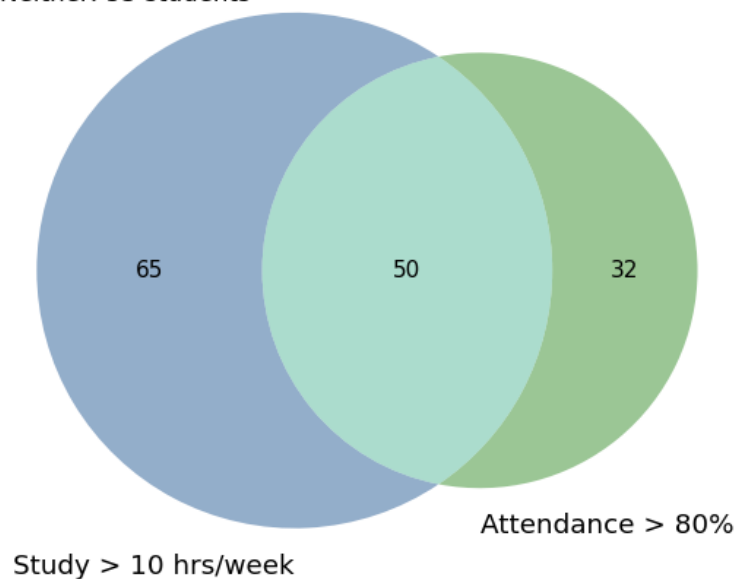
Task 4: Venn Diagram

Set A = students who study more than 10 hours per week. Set B = students who attend more than 80% of classes. The overlap A and B = students who satisfy both.

Region	Number of students
Only A (study > 10, attendance 80 or less)	65
Only B (attendance > 80, study 10 or less)	32
Both A and B (overlap)	50
Neither	53
Total	200

Venn diagram (total students = 200)

Neither: 53 students



$$P(A) = 115/200 = 0.575$$

$$P(B) = 82/200 = 0.410$$

$$P(A \text{ and } B) = 50/200 = 0.250$$

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B) = 0.575 + 0.410 - 0.250 = 0.735$$

Meaning: 50 students (25%) are strong in both study hours and attendance, while 53 students (26.5%) are in neither group.

Task 5: Contingency Table and Probability Calculations

Group discussion	Pass	Fail	Total
Yes	60	44	104
No	31	65	96
Total	91	109	200

Joint probability

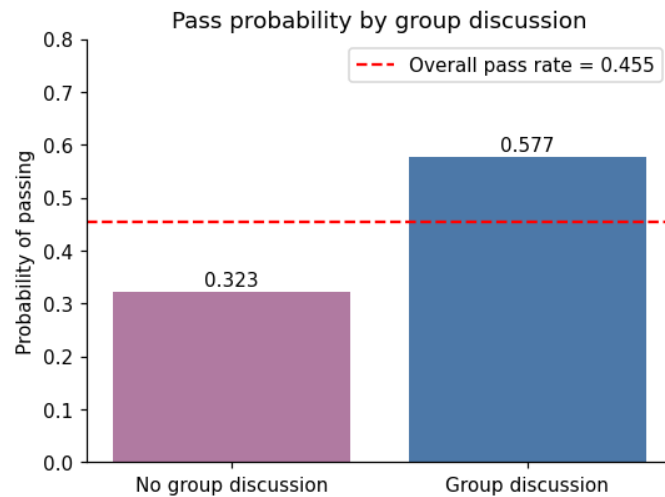
$$P(\text{GD} = \text{Yes and Pass}) = 60 / 200 = 0.3000$$

Marginal probability

$$P(\text{Pass}) = 91 / 200 = 0.4550$$

Conditional probability

$$\begin{aligned} P(\text{Pass} \mid \text{GD} = \text{Yes}) &= P(\text{Pass and GD}) / P(\text{GD}) \\ &= (60/200) / (104/200) = 0.3000 / 0.5200 = 0.5769 \end{aligned}$$



Task 6: Understanding Relationships

Intuition behind conditional probability

Conditional probability means **changing the sample space**. When we know a student joined group discussions, we stop looking at all 200 students and look only at the 104 students who joined. Then we ask: out of these, how many passed? The answer is $60/104 = 0.577$.

A student who joins group discussions has about a **57.7%** chance of passing, higher than the overall **45.5%**. So knowing about group discussion gives us useful extra information.

Independent, dependent or mutually exclusive?

Independent events: $P(A \text{ and } B) = P(A) \times P(B)$, or $P(A|B) = P(A)$.

Mutually exclusive events: $P(A \text{ and } B) = 0$.

Test	Calculation	Result
Independence	$P(\text{GD}) \times P(\text{Pass}) = 0.520 \times 0.455 = 0.2366$ vs $P(\text{GD and Pass}) = 0.3000$	Not equal, so NOT independent
Conditional check	$P(\text{Pass} \text{GD}) = 0.5769$ vs $P(\text{Pass}) = 0.4550$	Different, so dependent
Mutually exclusive	$P(\text{GD and Pass}) = 0.3000$	Not 0, so NOT mutually exclusive
Chi-square test	chi-square = 11.984, df = 1, p-value = 0.00054	$p < 0.05$, so dependent

Conclusion: Group discussion and passing are **dependent** events and they are **not mutually exclusive** (many students did both). Dependent only means they are linked in this data. It does not prove that group discussion causes passing.

Task 7: Bayes Theorem

Given: $P(H | \text{Pass}) = 0.70$, $P(H | \text{Fail}) = 0.40$, $P(H) = 0.60$, where H = high attendance (above 80%).

Find: $P(\text{Pass} | H)$.

$$\text{Bayes theorem: } P(\text{Pass} | H) = [P(H | \text{Pass}) \times P(\text{Pass})] / P(H)$$

$P(\text{Pass})$ is not given, so we find it with the law of total probability:

$$\begin{aligned} P(H) &= P(H|\text{Pass}) \times P(\text{Pass}) + P(H|\text{Fail}) \times [1 - P(\text{Pass})] \\ 0.60 &= 0.70x + 0.40(1 - x) \\ 0.60 &= 0.40 + 0.30x \\ x &= 0.20 / 0.30 = 0.6667 \end{aligned}$$

Now we put the values into Bayes theorem:

$$\begin{aligned} P(\text{Pass} | H) &= (0.70 \times 0.6667) / 0.60 \\ &= 0.4667 / 0.60 = 0.7778 \end{aligned}$$

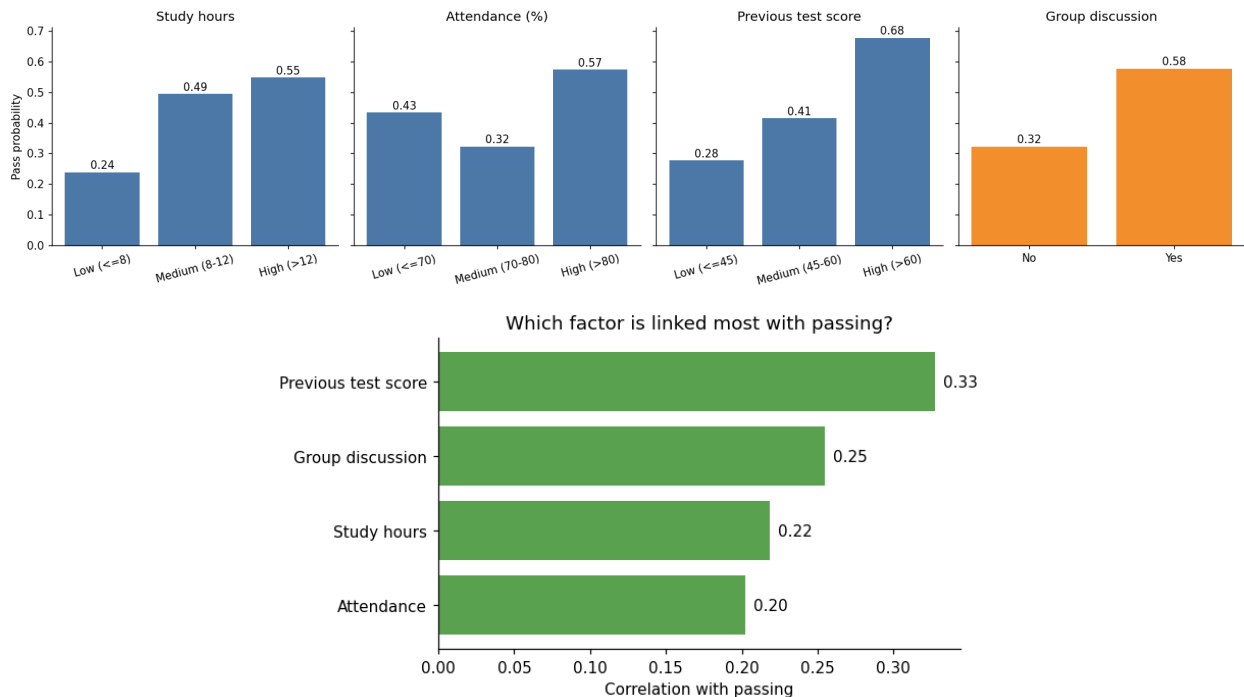
Answer: about 77.8%. A student with high attendance has a 77.8% chance of passing in this historical data.

Check: $P(\text{Fail} | H) = (0.40 \times 0.3333) / 0.60 = 0.2222$. Then $0.7778 + 0.2222 = 1$, so the answer is correct.

Note: These historical numbers imply a pass rate of 66.7%, while our sample of 200 students has 45.5%. They are different groups, which is fine because the question says to suppose this history.

Final Summary: What Affects the Chance of Passing?

Average	Passed	Failed
Study hours per week	11.72	10.18
Attendance (%)	80.56	75.79
Previous test score	59.58	49.16



Main findings

- The overall chance of passing is 45.5% (91 of 200 students).
- **Previous test score** has the strongest link with passing (correlation 0.33). Students who scored above 60 passed about 68% of the time, against 28% for scores of 45 or below.
- **Group discussion** is next (correlation 0.25). The pass chance is 57.7% with discussion and 32.3% without it.
- **Study hours** (0.22) and **attendance** (0.20) also help, but less. Students studying 8 hours or less passed only about 24% of the time.
- Group discussion and passing are dependent events, but this is a link and not proof of cause.
- **Limitation:** The data is AI-generated, so the results show the method and not real student behaviour.